

COURSE: CS 102 — Data Structures and Algorithms I

Credits: 3 | Department: Computer Science

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PREREQUISITES: CS 101 (minimum grade C)

CO-REQUISITES: None (CS 210 recommended to be taken concurrently)

OFFERED: Fall, Spring

DESCRIPTION:

Introduction to fundamental data structures and algorithms. Topics include arrays, linked lists, stacks, queues, trees (binary, BST, AVL), heaps, hash tables, and graphs. Algorithm analysis using Big-O notation. Sorting algorithms (merge sort, quicksort, heapsort). Students implement data structures in Java.

LEARNING OUTCOMES:

- Implement core data structures from scratch
- Analyze algorithm time and space complexity
- Select appropriate data structures for given problems
- Trace through and debug data structure operations

GRADING: Lab assignments 40%, Midterm 25%, Final 35%.

MINIMUM GRADE TO SATISFY PREREQUISITE FOR CS 201: C

NOTE: Students who received a C- in CS 101 must receive instructor consent before enrolling in CS 102. Contact the EECS undergraduate advisor.